

Project Documentation

Date	2 July 2026
Team ID	SWTID-2026-2260
Project Name	Smart Lender
Maximum Marks	3 Marks

Project Title

Smart Lender – AI Powered Loan Eligibility Prediction System

1. Introduction

Smart Lender is a Machine Learning-based web application that predicts whether a loan applicant is eligible for a loan. The application analyzes financial information such as monthly income, credit score, loan amount, and years of employment to provide instant loan eligibility predictions.

The project uses the XGBoost classification algorithm, which was selected as the best-performing model after comparing multiple machine learning algorithms. The application is developed using Python and Flask and provides a simple web interface for users to enter their details and receive a prediction.

2. Problem Statement

Banks and financial institutions receive a large number of loan applications every day. Manual evaluation of every application is time-consuming and may lead to inconsistent decisions.

This project aims to automate the initial loan screening process using Machine Learning, enabling faster, accurate, and data-driven loan eligibility predictions.

3. Objectives

- Develop a machine learning model for loan eligibility prediction.
- Compare multiple classification algorithms.
- Deploy the best model using Flask.
- Store applicant details in a database.
- Send prediction results to the applicant via email.
- Provide a simple and user-friendly interface.

4. Technologies Used

- Python

- Flask
- XGBoost
- Scikit-Learn
- Pandas
- NumPy
- HTML5
- CSS3
- SQLite
- Joblib
- Gmail SMTP

5. System Requirements

Hardware

- Intel i3 Processor or above
- 4 GB RAM (8 GB Recommended)
- 500 MB Free Storage

Software

- Windows 10/11
- Python 3.14
- Visual Studio Code
- Flask
- Jupyter Notebook

6. Dataset Description

The project uses a loan approval dataset containing applicant financial information.

Features

- Income
- Credit Score
- Loan Amount
- Years Employed

Target Variable

- Loan Approved
- Loan Rejected

7. Methodology

The project follows these steps:

1. Dataset Collection
2. Data Cleaning
3. Exploratory Data Analysis
4. Feature Selection
5. Model Training
6. Model Evaluation
7. Model Saving
8. Flask Application Development
9. Prediction
10. Database Storage
11. Email Notification

8. Machine Learning Models

The following algorithms were trained and evaluated:

- Decision Tree
- Random Forest
- K-Nearest Neighbors (KNN)
- XGBoost

Among these, **XGBoost** achieved the highest performance and was selected for deployment.

9. System Architecture

User

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Web Application (Flask)

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Input Validation

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XGBoost Model

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Prediction

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|—————▶ SQLite Database

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|—————▶ Email Notification

|



Prediction Result

10. Project Structure

SMART LENDAR/

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|— .venv/

|— Dataset/

| └─ loan_approval.csv

|— eda_plots/

|— flask/

| └─ static\css/

| └─ templates/

| └─ .env

|— app1.py

|— loan_model.pkl

|— .gitignore

|— edaanalysis.py

```
|— loan_model.pkl
|— loans.db
|— modelbuilding.py
|— preprocess.py
|— test.py
└— train_model.py
```

11. Features

- AI-Based Loan Prediction
- Real-Time Prediction
- SQLite Database Integration
- Email Notification
- User-Friendly Interface
- Secure Environment Variables
- Responsive Web Design

12. Results

The application successfully predicts loan eligibility using the trained XGBoost model. User details are stored in the SQLite database, and prediction results are sent automatically to the registered email address.

Smart Lender

AI-Powered Loan Approval Prediction System Architecture

Evaluate commercial loan underwriting criteria instantly using optimized machine learning modeling layers.

[Launch Evaluation Form](#)

XGBoost Engine

Leverages high-performance gradient boosted decision trees for real-time risk assessment parameters and continuous feature importance auditing.

Automated Logging

Every parsed user submission profile dynamically tracks directly to transaction relational storage files for offline record keeping and compliance checks.

Instant Notification

Dispatches automated status updates and complete data profile summaries back to client communication layers instantly via custom server mailing channels.

Loan Application Assessment

Fill out the form fields completely to run the predictive qualification layer.

Full Name	Email Address	
John Doe	name@example.com	
Annual Income (₹)	Credit Score (300 - 850)	
e.g. 65000	e.g. 720	
Requested Loan Amount (₹)	Years Employed	Risk Points (0-100)
e.g. 30000	e.g. 4	e.g. 45.5
<button>Evaluate Eligibility Profile</button>		

[— Return to Home Screen](#)

SYSTEM ANALYSIS RESULT

Sorry divya, your Loan application has been Denied.

DECISION BREAKDOWN:

This assessment was reached primarily due to a low credit score profile.

An automated breakdown details summary and dynamic compliance confirmation log file copies have been dispatched via network transmission routing paths directly to the email address provided.

[Evaluate Another Application](#)

[Exit Dashboard Platform](#)

SYSTEM ANALYSIS RESULT

Congratulations sirisha! Your Loan is Approved.

DECISION BREAKDOWN:

This assessment was reached supported primarily by an excellent income-to-loan safety ratio.

An automated breakdown details summary and dynamic compliance confirmation log file copies have been dispatched via network transmission routing paths directly to the email address provided.

[Evaluate Another Application](#)

[Exit Dashboard Platform](#)

13. Future Enhancements

- Cloud Deployment (IBM Cloud/AWS/Azure)
- User Authentication
- Admin Dashboard
- Loan History Tracking
- PDF Report Generation
- Advanced Analytics Dashboard

14. Conclusion

The Smart Lender application demonstrates how Machine Learning can support financial institutions by automating loan eligibility prediction. By integrating XGBoost, Flask, SQLite, and email notifications, the project provides a complete end-to-end solution that is efficient, scalable, and user-friendly.